CSCI E-106:Assignment 2

Due Date: February 13, 2026 at 11:59 pm EST

Problem 1

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X.

```
cdi <- read.csv("CDI Data.csv")
summary(cdi)
```

```
##  Identification.number    County              State              Land.area
##  Min.   :  1.0         Length:440         Length:440         Min.   :   15.0
##  1st Qu.:110.8         Class :character   Class :character   1st Qu.:  451.2
##  Median :220.5         Mode  :character   Mode  :character   Median :  656.5
##  Mean   :220.5                                               Mean   : 1041.4
##  3rd Qu.:330.2                                               3rd Qu.:  946.8
##  Max.   :440.0                                               Max.   :20062.0
##  Total.population  Percent.of.population.aged.18.34
##  Min.   : 100043   Min.   :16.40
##  1st Qu.: 139027   1st Qu.:26.20
##  Median : 217280   Median :28.10
##  Mean   : 393011   Mean   :28.57
##  3rd Qu.: 436064   3rd Qu.:30.02
##  Max.   :8863164   Max.   :49.70
##  Percent.of.population.65.or.older Number.of.active.physicians
##  Min.   : 3.000                    Min.   :   39.0
##  1st Qu.: 9.875                    1st Qu.:  182.8
##  Median :11.750                    Median :  401.0
##  Mean   :12.170                    Mean   :  988.0
##  3rd Qu.:13.625                    3rd Qu.: 1036.0
##  Max.   :33.800                    Max.   :23677.0
##  Number.of.hospital.beds Total.serious.crimes Percent.high.school.graduates
##  Min.   :   92.0         Min.   :   563       Min.   :46.60
##  1st Qu.:  390.8         1st Qu.:  6220       1st Qu.:73.88
##  Median :  755.0         Median : 11820       Median :77.70
##  Mean   : 1458.6         Mean   : 27112       Mean   :77.56
##  3rd Qu.: 1575.8         3rd Qu.: 26280       3rd Qu.:82.40
##  Max.   :27700.0         Max.   :688936       Max.   :92.90
##  Percent.bachelor.s.degrees Percent.below.poverty.level Percent.unemployment
##  Min.   : 8.10              Min.   : 1.400              Min.   : 2.200
##  1st Qu.:15.28              1st Qu.: 5.300              1st Qu.: 5.100
##  Median :19.70              Median : 7.900              Median : 6.200
##  Mean   :21.08              Mean   : 8.721              Mean   : 6.597
##  3rd Qu.:25.32              3rd Qu.:10.900              3rd Qu.: 7.500
##  Max.   :52.30              Max.   :36.300              Max.   :21.300
##  Per.capita.income Total.personal.income Geographic.region
##  Min.   : 8899     Min.   :  1141        Min.   :1.000
##  1st Qu.:16118     1st Qu.:  2311        1st Qu.:2.000
##  Median :17759     Median :  3857        Median :3.000
```

```
##  Mean   :18561     Mean   :  7869        Mean   :2.461
##  3rd Qu.:20270     3rd Qu.:  8654        3rd Qu.:3.000
##  Max.   :37541     Max.   :184230        Max.   :4.000
```

```
variables = names(cdi)
numeric_variables = names(cdi[, sapply(cdi, is.numeric)])
response_variable = "Number.of.active.physicians"
explanatory_variable = "Total.personal.income"
```

a-) Built a regression model to predict Y by using X. Write down the stand error of the slope and intercept.

```
response_variable = "Number.of.active.physicians"
explanatory_variable = "Total.personal.income"

fit = lm(Number.of.active.physicians ~ Total.personal.income, cdi)
summary(fit)
```

```
##
## Call:
## lm(formula = Number.of.active.physicians ~ Total.personal.income,
##     data = cdi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1926.6  -194.5   -66.6    44.2  3819.0
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -48.39485   31.83333   -1.52    0.129
## Total.personal.income   0.13170    0.00211   62.41   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 569.7 on 438 degrees of freedom
## Multiple R-squared:  0.8989, Adjusted R-squared:  0.8987
## F-statistic:  3895 on 1 and 438 DF,  p-value: < 2.2e-16
```

b-) Are the slope and intercept statistically significant? Write down your null and alternative hypothesis and p value of the test. Calculate the 95% Confidence interval for the slope. (20 points)

c-) Predict the number of active physicians for total personal income of 100000,250000 and 50000. Calculate the 95% confidence interval (Hint: Use the range of total personel income to identify which observations are within the range of the training data) (20 points)

d-) What is the error variance of the regression line?(5 points)

Problem 2

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X. Select 1,000 random samples of 400 observations, fit the regression model and record β_0 and β_1 for each selected sample. Calculate the mean and variance of β_0 and β_1 based on the 1,000 different regression line and compare against the regression model in question 1 part b. (50 points)